

- 1 A student is determining the acceleration of free fall g using a pendulum. Fig. 1.1 shows the pendulum. Fig. 1.2 shows one complete oscillation of the pendulum.

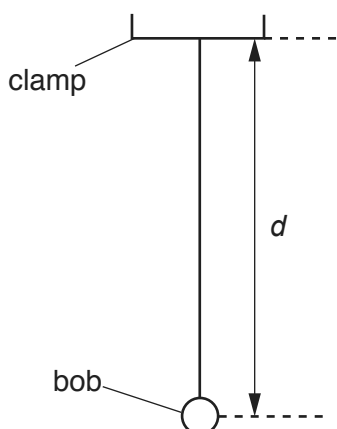


Fig. 1.1

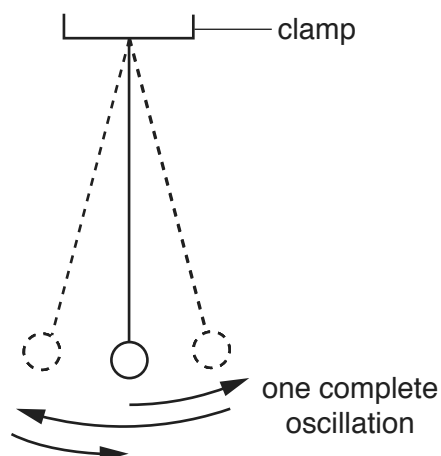


Fig. 1.2

- (a) On Fig. 1.1, measure the distance d .

$d =$ cm [1]

- (b) Fig. 1.1 is drawn $1/10^{\text{th}}$ actual size.

- (i) Calculate the actual distance D from the bottom of the clamp to the centre of the bob.

$D =$ cm [1]

The student displaces the bob slightly and releases it so that it swings. He measures the time t for 10 complete oscillations. The time t is shown on the stopwatch in Fig. 1.3.



Fig. 1.3

- (ii) Write down the time t shown in Fig. 1.3.

$t =$ [1]

- (iii) Calculate the period T of the pendulum. The period is the time for one complete oscillation.

$T =$ [1]

- (iv) Calculate T^2 .

$$T^2 = \dots\dots\dots [1]$$

- (v) Calculate the acceleration of free fall g using the equation $g = \frac{20}{T^2}$.

$$g = \dots\dots\dots [1]$$

- (c) The student adjusts the pendulum until the distance D measured to the centre of the bob is 100.0 cm.

He repeats the procedure and obtains another value of T^2 .

$$T^2 = \dots\dots\dots 3.94$$

- (i) On the dotted line above, write the unit for T^2 . [1]

- (ii) Calculate the acceleration of free fall g using the equation $g = \frac{40}{T^2}$ and the value of T^2 from (c). Give your answer to a suitable number of significant figures for this experiment.

$$g = \dots\dots\dots [1]$$

- (d) Another student states that repeating the experiment improves the reliability of the value obtained for g .

Suggest **two** changes that you would make to improve the reliability. The stopwatch cannot be changed.

1.

.....

2.

.....

[2]

- (e) State **one** precaution that you would take in this experiment in order to obtain accurate readings.

.....

..... [1]

[Total: 11]

2 A student is investigating resistance.

She uses the circuit shown in Fig. 2.1.

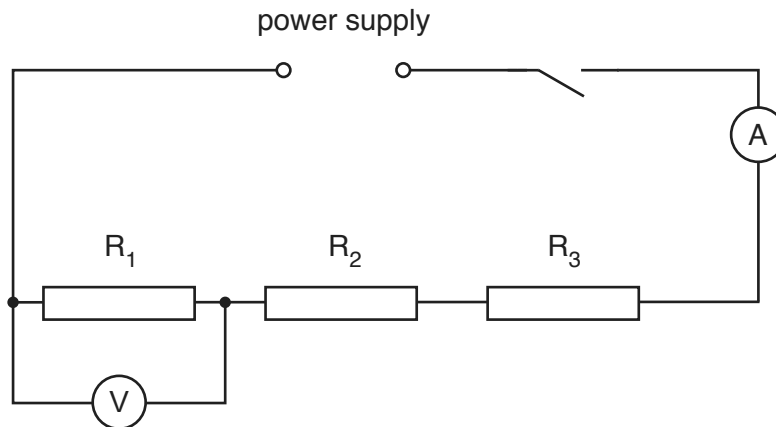


Fig. 2.1

- (a) She measures the potential difference V_1 across the resistor R_1 and the current I in the circuit.

Figs. 2.2 and 2.3 show the voltmeter and ammeter readings.

- (i) Write down the readings shown on the meters in Figs. 2.2 and 2.3.

$V_1 = \dots\dots\dots$

$I = \dots\dots\dots$

[2]

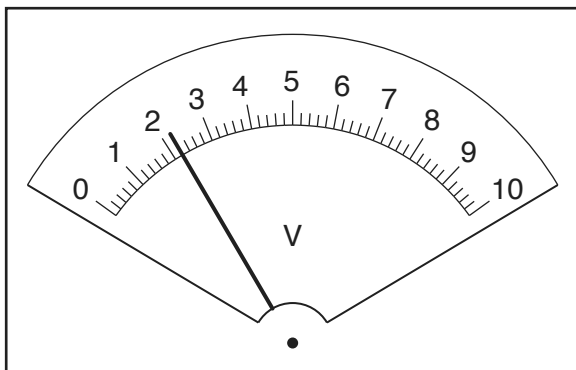


Fig. 2.2

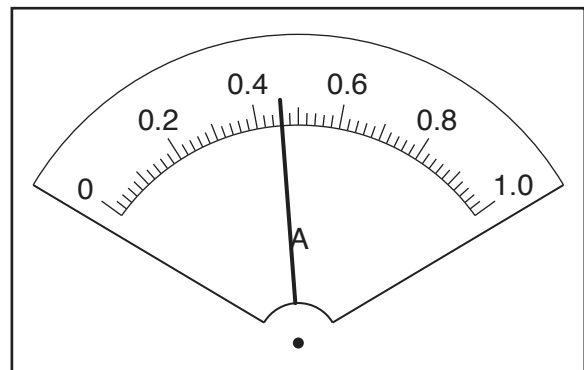


Fig. 2.3

- (ii) Calculate the resistance of the resistor R_1 using the equation $R_1 = \frac{V_1}{I}$.

$$R_1 = \dots\dots\dots [1]$$

- (b) The student measures the potential difference across each resistor in turn. She calculates values for the resistance R_2 and R_3 of the resistor R_2 and of the resistor R_3 . She records her values as:

$$R_2 = \dots\dots\dots 4.75 \Omega$$

$$R_3 = \dots\dots\dots 4.81 \Omega$$

State whether the results suggest that the three resistors have the same value of resistance. Justify your statement by reference to the results.

statement

justification

.....

.....

[2]

- (c) Calculate the combined resistance R of resistors R_1 , R_2 and R_3 connected in series, using the equation $R = R_1 + R_2 + R_3$. Give your answer to a suitable number of significant figures for this experiment.

$$R = \dots\dots\dots [2]$$

- (d) The student checks her result by connecting the voltmeter across all three resistors connected in series. Tick the potential difference reading you would expect to be closest to the reading she obtains.

☐	$\frac{V_1}{3}$
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☐	V_1
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☐	$3V_1$
--------------------------	--------

[1]

(e) Complete the circuit diagram in Fig. 2.4 to show

- the three resistors connected in parallel
- the voltmeter connected to measure the potential difference across the resistors
- a variable resistor connected to control the current in all three resistors.

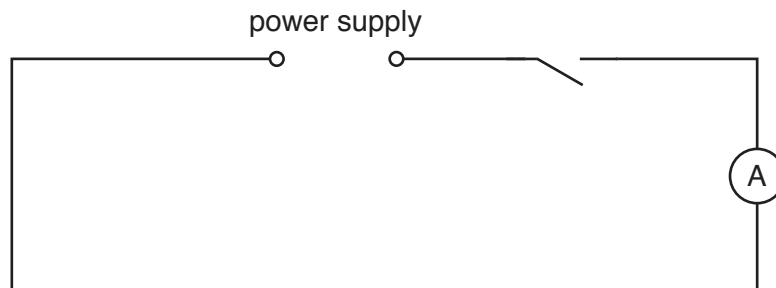


Fig. 2.4

[3]

[Total: 11]

Question 3 starts on the next page.

- 3 A student is determining the focal length f of a lens.

Fig. 3.1 shows the apparatus.

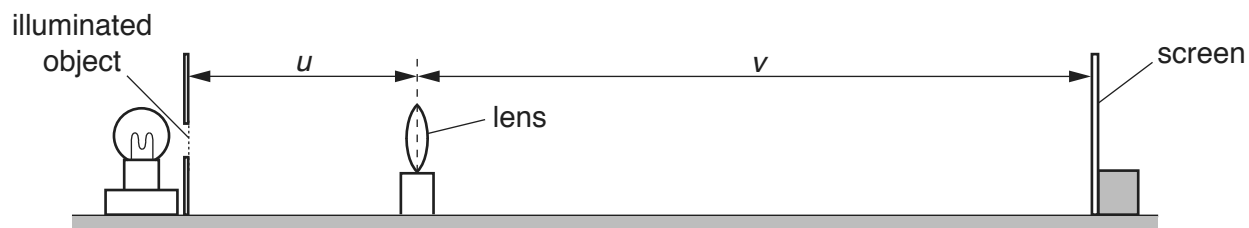


Fig. 3.1

- (a) The student places the screen a distance $D = 70.0$ cm from the illuminated object.

He places the lens close to the screen and moves the lens slowly away from the screen until a clearly focused image is formed on the screen.

He measures the distance u between the centre of the lens and the illuminated object.

He measures the distance v between the centre of the lens and the screen.

He repeats the procedure using values for D of 75.0 cm, 80.0 cm, 85.0 cm and 90.0 cm.

The readings are shown in Table 3.1.

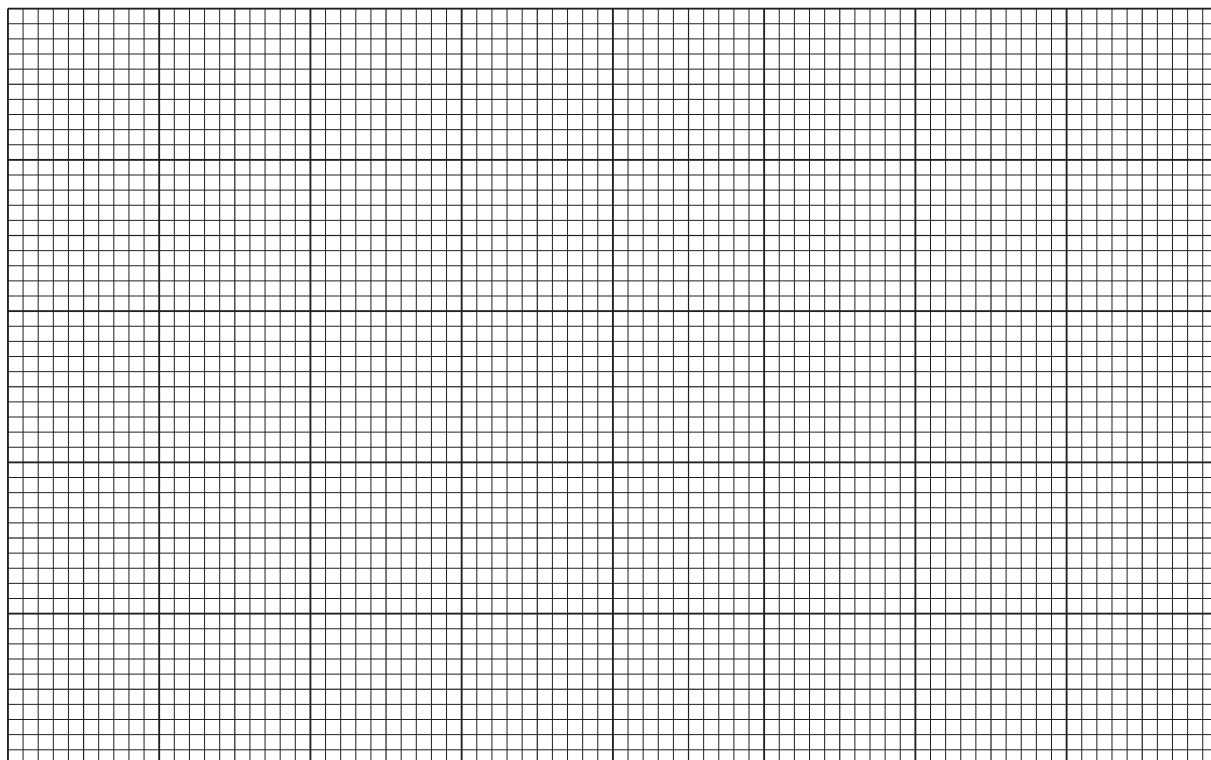
Calculate, and record in Table 3.1, uv for each value of D .

Table 3.1

D/cm	u/cm	v/cm	uv/cm^2
70.0	22.0	48.4	
75.0	20.7	54.5	
80.0	20.0	60.0	
85.0	19.5	65.8	
90.0	19.0	71.2	

[1]

- (b) Plot a graph of uv/cm^2 (y -axis) against D/cm (x -axis). You do **not** need to start your axes at the origin (0,0).



[4]

- (c) Determine the gradient G of the line. Show clearly on the graph how you obtained the necessary information.

$G =$ [2]

- (d) The focal length f of the lens is numerically equal to the gradient G of the graph. Write down a value for the focal length f of the lens. Give your answer to a suitable number of significant figures for this experiment.

$f =$ [2]

- (e) Suggest **two** difficulties in this experiment when trying to obtain accurate readings.

1.

 2.

[2]

[Total: 11]

- 4 A student is investigating the effect of double-walled insulation on the rate of cooling of hot water in a copper container. The student places the copper container inside a larger metal container. He is investigating the effect of the size of the air gap between the copper container and larger metal containers.

Plan an experiment to investigate the effect of the size of the air gap between the copper container and larger metal containers on the rate of cooling of hot water.

The following apparatus is available:

- a copper container
- a number of metal containers of different diameters (all larger than the copper container)
- a thermometer
- a stopwatch
- a measuring cylinder
- a supply of hot water.

You can also use other apparatus and materials that are usually available in a school laboratory.

In your plan, you should:

- explain briefly how you would carry out the investigation
- state the key variables that you would control
- draw a table, or tables, with column headings, to show how you would display your readings (you are not required to enter any readings in the table)
- explain how you would use your readings to reach a conclusion.

You may draw a diagram if it helps your explanation.

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